

Element A student work

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The turtle tank currently doesn't provide a very efficient or sustainable system for maintaining clean and healthy water conditions. The tank holds up to 65.6 gallons, but is typically filled to 55.1 gallons. It requires a full drain and refill every 3-4 weeks, which takes about 2 hours total. This indicates that the current filtration and maintenance setup isn't adequately maintaining water quality over time. As a result, there is a need for a more efficient system that reduces maintenance time while keeping up with proper living conditions. The current basking platform does not provide enough space for both turtles to use it at the same time. The platform measures 15.5 × 13 × 10 inches, while the female turtle alone is approximately 10–11 inches long. This creates a situation where both turtles cannot comfortably fit on the platform simultaneously, limiting access to essential basking space. The problem involves maintaining water quality in a 55.1-gallon turtle and that houses two turtles, The female, Bella, measuring up to 10-11 inches long, and the male, Zark being about 6 inches long. Turtles produce a significant amount of waste compared to fish, which leads to faster water contamination.

The tank uses an AquaClear power filter, which processes about 1.33 gallons per minute and includes mechanical, chemical, and biological filtration systems. Despite this, the tank still requires pretty frequent full cleanings every 3-4 weeks, suggesting that the filtration system may not be sufficient for the tank size and bio load. Additionally the tank includes a basking platform that's 15.5 × 13 × 10 inches, a 13-watt halogen bulb that simulates daylight and a 5-watt UVB bulb for turtle health.

The issue occurs continuously as waste builds up over time, leading to poor water quality. This affects both the turtle's health and the time/effort required by the person maintaining the tank. The problem is significant because insufficient filtration leads to frequent maintenance, which is time consuming and may not fully prevent unhealthy conditions.

Basking is essential for turtles because it allows them to regulate body temperature and absorb UVB light, which supports shell and bone health. In this tank, there are two turtles of significantly different sizes, with the larger female taking up most of the basking area. Because the platform isn't large enough, Bella often occupies the entire space, Zark may be pushed off or prevented from basking, and both turtles cant bask at the same time. This issue occurs daily and repeatedly, especially when both turtles attempt to bask under the heat and UVB lamps. The problem is not just about space, but also about access to necessary environmental conditions.

This problem connects to broader issues of animal care, efficiency, and sustainability. Poor water quality can lead to health problems for turtles, including infections or stress. It also impacts the tank owner, who must spend multiple hours every few weeks performing maintenance.

In other contexts, similar problems exist in aquariums, zoos, and aquatic research facilities, where filtration and water quality are critical. Solving this problem could lead to reduced maintenance time, improved animal health, and more efficient use of resources like water and energy. This problem also highlights the balance between system design and biological needs , which is an

important concept in engineering and environmental management. This problem relates to animal behavior and habitat design. In many shared animal environments, limited space can lead to competition or dominance behaviors, as seen with the female turtle.

If not addressed, this can lead to unequal access to heat and UVB light, increased stress or aggression between turtles, potential long-term health issues for the turtle that cannot bask regularly.

Similar issues occur in zoos, aquariums, and pet enclosures, where space and resource access must be carefully designed to accommodate multiple animals.

The several primary stakeholders that are affected are the turtles, the tank owner/caretaker, others who occupy the lab, and the equipment as a part of the system. Due to waste buildup and inconsistent water quality, there is a need to improve filtration to ensure a healthy living environment. For the caretaker, due to the 2-hour cleaning process, there is a need to reduce maintenance time and effort. Any potential odors or mess from frequent cleaning may affect the other people who occupy the lab. And, because of the limitations in processing waste effectively, there is a need to evaluate or improve filtration performance.

The measurements provided throughout show that the filtration system may not be keeping up with the waste produced, maintenance is time-intensive, and the current setup is inefficient for long-term use. This data all demonstrates a clear need for improvement in system design or efficiency.

The need to solve this problem is important because it affects both animal health and system efficiency. The current system requires frequent, time-consuming maintenance despite having a functioning filter. With over 55 gallons of water needing replacement every few weeks, this also results in significant water usage.

By improving the system it would be possible to maintain cleaner water for longer periods, reduce the 2-hour maintenance time, improve the health and well-being of the turtles, as well as increasing the overall efficiency of the tank system.

Solving the problem of the basking platform is important because it is essential to turtle health.

Without consistent access to heat and UVB light, turtles may develop health issues over time.

The current platform design limits accessibility and creates competition, which reduces the effectiveness of the habitat. By improving the size or structure of the basking area, both turtles could safely and consistently meet their biological needs.

Did you want to focus on water quality or basking? You have taken on a number of topics here.

Also, any tank would require filtration and frequent cleanings. You would never simply filter the water and not clean the tank. I don't know if that is a fair criticism.